

PROJECT AND TEAM INFORMATION

Project Title

LogX : Intelligent Log Analyzer

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Servers and applications are used on large scale everyday. and as we know that these generate a large amount of log data. for a instance if a small project contains fewer logs like 50-100, they can be checked manually. But what if they are in thousands or millions, no one would check to see which one contain the most errors and how much database errors are there. this is very time consuming. Our project solves this problem by automatically analyzing log files and find the useful information and patterns from them by using DSA and OOP's concepts. The user will create a folder which would contain the log files. The application would read and then process those files. then the user can search and analyze among the logs using keywords. To find errors in the application, check error frequency and pattern identification there would be different options. And to implement these features concepts like: Hash map, Priority Queue, Trie, Dynamic Programming and sliding window will be used.

State of the Art / Current solution (1 pt)

Nowadays, applications like text editor, command line tools and different log viewer tools are used to analyze log files. The user can search, filter and analyze log files by using these tools. But the problem is that all these features can't be there in one specific application. the user may need to switch between different applications in order to use them. this can be frustrating. our idea is to make searching, Indexing and basic analysis in one single application. it would make index of content of the log files, so that the user can easily search the log files. Then the user can generate the report according to time and pattern.

Project Goals and Milestones (2 pts)

The main goal of our project is to make an intelligent log analyzer which will help the user to easily load, search, organize and analyze the log files.

initial milestones are as follows:

- 1.Loading and storing log files.*
- 2.Making a basic structure/index by processing the logs.*
- 3.Implementing basic search, sorting, and time range search.*
- 4.Adding Auto complete and spelling correction feature.*
- 5.Making a report after the analysis.*

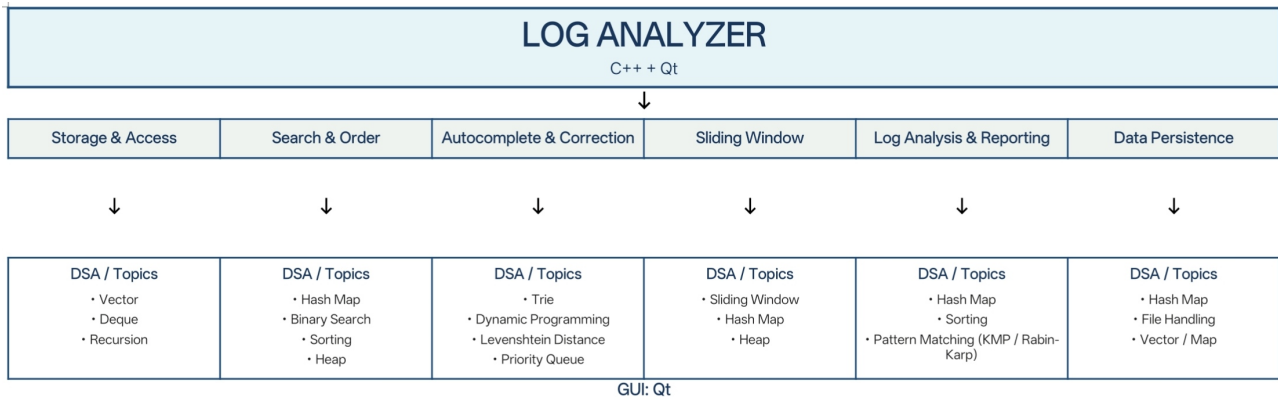
Further milestones are as follows:

- 1.Detecting frequent errors or error spikes.*
- 2.Pattern matching or log analysis.*
- 3.Generating result on the basis of analysis.*
- 4.Saving the processed data.*
- 5.Adding all the features in QT GUI to make a complete application.*

Project Approach (3 pts)

We are trying to make a simple log analyzer. which basic work is to load the log files and save the entry of every log file. Especially it's timestamp, type, keyword,etc... The project would be made using C++ language. the application will first ask the user to select the log folder which contains the log files. And once the folder is selected it would analyze all the files inside and store the data. Hash map is used to search or count the errors. Trie is used for features like auto complete. Priority Queue is used to rank the result on the basis of frequent and important errors. Dynamic programming with Edit distance is used for spelling suggestions. And sliding window is used to detect increase/decrease in errors according to the time. we will make GUI using QT c++ library which will include adding folder button, search bar, and options to access application features. to organize the code we will use CMake, and Github for version control. At first the application will be made for windows platform and for development windows and C++ compiler will be used.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The main outcome of this C++ based GUI application is to easily load log files and analyze them. Some main deliverables are: 1- The application provides a log files loading and processing system to store and process data properly. 2- A search system to search keywords or errors quickly. 3- Autocomplete and autocorrection feature to help users in searching. 4- Log analysis features to identify frequent errors, important errors, and time-based errors. 5- Reports and results to present the analyzed log data in a simple and efficient way. 6- QT-based GUI to help users access the features of the application without using complicated commands. 7- Implementation of DSA and OOP in which concepts like hash map, trie, heap, binary search and sliding window can be used practically. Finally, with the completion of the project, report will be deliver in which system design, algorithms, implementation and results will be explained.

Assumptions

- 1- Log files would be in proper format and they should contain timestamp, log type, message etc.
- 2- The application would be able to access the logs.
- 3- User should know how to do the basic search.
- 4- To process large log files efficiently, DSA-based methods would be used.
- 5- Log files should contain valid data as system will mainly focus on analyzing valid log data.

References

*Books:**Data Structures and Algorithms in C by E. Balagurusamy**Object-Oriented Programming in C++ by Mark Allen Weiss**Concept References:**GeeksforGeeks – Trie Data Structure**GeeksforGeeks – Binary search**GeeksforGeeks – Edit Distance (DP)**QT Documentation:**Gui design, buttons, text fields and application interface develop.*